MTPro2 Math specimens

1 Three math sizes and compound expressions

Compare expressions in normal, superscript and nested superscript positions.

Inline example: $x^n + y^n = z^n$.

$$\sum_{i=1}^{n} \sin x + i^{\sin x} + i^{i^{\sin x}}$$

$$A,\ldots,Z \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$2^{A,\ldots,Z} \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$2^{2^{A,\ldots,Z}} \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$\aleph_\alpha \times \aleph_\beta = \beta \Leftrightarrow \alpha \leq \beta$$

$$2^{\aleph_\alpha \times \aleph_\beta = \beta \Leftrightarrow \alpha \leq \beta}$$

$$2^{2^{\aleph_\alpha \times \aleph_\beta = \beta \Leftrightarrow \alpha \leq \beta}}$$

$$\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon$$

$$2^{\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon}$$

$$2^{2^{\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon}}$$

$$|x - a| < \delta \Rightarrow |f(x) - l| < \varepsilon$$

$$2^{|x-a|<\delta \Rightarrow |f(x)-l|<\varepsilon}$$

$$2^{2^{|x-a|<\delta \Rightarrow |f(x)-l|<\varepsilon}}$$

$$\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}$$

$$2^{\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}}$$

$$2^{2^{\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}}}$$

$$\{x | x \neq x\} = \emptyset \qquad (A \cap B)^\circ \subset A^\circ \cap B^\circ$$

$$2^{\{x|x\neq x\}=\emptyset} \qquad (A\cap B)^\circ \subset A^\circ \cap B^\circ$$

$$2^{2^{\{x|x\neq x\}=\emptyset}} \qquad (A\cap B)^\circ \subset A^\circ \cap B^\circ$$

$$\omega = \nu + v(x, y)\,dx + w(x, y)\,dy + d\varkappa$$

$$2^{\omega=\nu+v(x,y)\,dx+w(x,y)\,dy+d\varkappa}$$

$$2^{2^{\omega=\nu+v(x,y)\,dx+w(x,y)\,dy+d\varkappa}}$$

$$d\omega = dv + \left(\frac{\partial w}{\partial x} - \frac{\partial v}{\partial y}\right) dx \wedge dy$$

$$2^{d\omega=dv+\left(\frac{\partial w}{\partial x}-\frac{\partial v}{\partial y}\right)dx\wedge dy}$$

$$2^{2^{d\omega=dv+\left(\frac{\partial w}{\partial x}-\frac{\partial v}{\partial y}\right)dx\wedge dy}}$$

$$\hat{x} + \hat{X} + \widehat{xy} + \widehat{xyz} + \vec{A}$$

$$2^{\hat{x}+\hat{X}+\widehat{xy}+\widehat{xyz}+\vec{A}}$$

$$2^{2^{\hat{x}+\hat{X}+\widehat{xy}+\widehat{xyz}+\vec{A}}}$$

$$R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij}$$

$$2^{R_{ijkl}=-R_{jikl}=-R_{ijlk}=R_{klij}}$$

$$2^{2^{R_{ijkl}=-R_{jikl}=-R_{ijlk}=R_{klij}}}$$

$$(f \circ g)'(x) = f'\big(g(x)\big) \cdot g'(x)$$

$$2^{(f\circ g)'(x)=f'(g(x))\cdot g'(x)}$$

$$2^{2^{(f\circ g)'(x)=f'(g(x))\cdot g'(x)}}$$

$$f(x) = \begin{cases} |x| & x > a \\ -|x| & x \le a \end{cases}$$

$$2^{f(x)=\begin{cases}|x| & x>a\\ -|x| & x\le a\end{cases}}$$

$$2^{2^{f(x)=\begin{cases}|x| & x>a\\ -|x| & x\le a\end{cases}}}$$

$$\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx = \sqrt{\pi}$$

$$2^{\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx=\sqrt{\pi}}$$

$$2^{2^{\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx=\sqrt{\pi}}}$$

$$X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j x^j \frac{\partial}{\partial \dot{x}^j}$$

$$2^{X=\sum_i \xi^i \frac{\partial}{\partial x^i}+\sum_j x^j \frac{\partial}{\partial \dot{x}^j}}$$

$$2^2 \quad X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j x^j \frac{\partial}{\partial \dot{x}^j}$$

2 Standard accents

Compare ordinary accents with wide accents over longer expressions.

$$\mathring{x}$$

$$\dot{\Gamma} + \ddot{\Gamma}$$

$$\begin{array}{llllll} \hat{A}, & \widehat{M} & & & & \\ \hat{A}, & \widehat{M}, & \widehat{xy}, & \widehat{xyz}, & \widehat{xyzw}, & \widehat{x+y+z+\cdots+w} \end{array}$$

$$\begin{array}{llllll} \tilde{A}, & \widetilde{M} & & & & \\ \tilde{A}, & \widetilde{M}, & \widetilde{xy}, & \widetilde{xyz}, & \widetilde{xyzw}, & \widetilde{x+y+z+\cdots+w} \end{array}$$

$$\begin{array}{llllll} \check{A}, & \check{M} & & & & \\ \check{A}, & \check{M}, & \check{xy}, & \check{xyz}, & \check{xyzw}, & \check{x+y+z+\cdots+w} \end{array}$$

$$\bar{M}, \qquad \overline{M}, \qquad \overline{x+y+z}$$

3 Math alphabets

Compare alphabet styles and letters in subscript and superscript positions.

$$A_{\mathbf{X}}(f) = \mathbf{X}(\mathbf{f}) = 2^{2^{\mathbf{X}(\mathbf{g})}}$$

ABGMXYZ
$ABGMXYZ$
ABGMXYZ
ABGMXYZ
$\mathcal{ABGMXYZ}$
$\boldsymbol{\mathcal{ABGMXYZ}}$
$\mathfrak{ABGMXYZ}$
$\boldsymbol{\mathfrak{ABGMXYZ}}$
$\mathbb{ABGMXYZ}$

4 Parentheses at four matrix sizes

$$\hat{J}_x^{(1/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \hat{J}_x^{(3/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

$$\hat{J}_x^{(5/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{5} & 0 & 0 & 0 & 0 \\ \sqrt{5} & 0 & 2\sqrt{2} & 0 & 0 & 0 \\ 0 & 2\sqrt{2} & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 & 2\sqrt{2} & 0 & \sqrt{5} \\ 0 & 0 & 0 & 0 & \sqrt{5} & 0 \end{pmatrix}$$

$$\hat{J}_x^{(7/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{7} & 0 & 2\sqrt{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 2\sqrt{3} & 0 & \sqrt{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{15} & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & \sqrt{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{15} & 0 & 2\sqrt{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2\sqrt{3} & 0 & \sqrt{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{7} & 0 \end{pmatrix}$$

5 Formulas from physics and mathematics

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} &= 0, \qquad p_i = \frac{\partial L}{\partial \dot{q}_i}, \\ \dot{q}_i &= \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \end{aligned}$$

$$\begin{cases} \nabla \cdot \boldsymbol{E} &= \dfrac{\rho}{\varepsilon_0}, \\ \nabla \cdot \boldsymbol{B} &= 0, \\ \nabla \times \boldsymbol{E} &= -\dfrac{\partial \boldsymbol{B}}{\partial t}, \\ \nabla \times \boldsymbol{B} &= \mu_0 \boldsymbol{j} + \mu_0 \varepsilon_0 \dfrac{\partial \boldsymbol{E}}{\partial t}. \end{cases}$$

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{r}, t) \right] \psi.$$

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{Z}, \qquad Z = \operatorname{tr}\left(e^{-\beta \hat{H}} \right), \qquad \beta = \frac{1}{k_{\mathrm{B}} T}.$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

$$\begin{aligned} \rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} \right) &= -\nabla p + \mu \nabla^2 \boldsymbol{u} + \rho \boldsymbol{f}, \\ \nabla \cdot \boldsymbol{u} &= 0. \end{aligned}$$

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \quad \Rightarrow \quad \mathbb{P}\left(\limsup_{n \to \infty} A_n \right) = \mathbb{P}\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n \right) = 0.$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k}^2 = \begin{cases} (-1)^{n/2} \dbinom{n}{n/2}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$